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Forecasting Inflation in a Big Data Setting Using Neural Networks

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1 Introduction

Forecasting inflation is essential for decision makings by firms, investors, and central banks. Time-series econometrics typically adopts linear univariate and multivariate models, such as autoregressive model (AR) and vector autoregression (VAR), to forecast inflation. Although these models are interpretable and computationally efficient, they face two limitations. First, linear models often fail to identify complex non-linear inflation dynamics. Second, linear models tend to over-fit in a high dimensional setting. Recent advances in machine learning (ML) provide tools to tackle these issues (Masini et al., 2023). Non-linear and non-parametric ML methods, such as random forest and neural networks, can provide accurate and robust predictions in high-dimensional setting. This research applies neural networks, including feed-forward neural networks (FNN) and long-short term memory networks (LSTM), to forecast US CPI inflation.

2 Literature Review

Inflation forecasting methods fall into two categories: univariate and multivariate. Univariate models take past inflation rates as predictors, while multivariate models incorporates additional covariates. Although using less information, univariate models initially proved hard to beat (Stock and Watson 2010). This result was largely because inflation after the 1980s became less volatile and more persistent, favoring univariate models (Stock & Watson, 2007). Another reason is that early multivariate models focused on a limited set of covariates (Medeiros et al. 2021).

With advances in ML and large macroeconomic datasets, multivariate models began to make systematic gains in forecasting accuracy. Medeiros and Mendes (2016) found that LASSO achieves superior performance in US CPI inflation forecasting relative to AR. Medeiros et al. (2021) extended the analysis to a broader set of ML methods. They found that random forests outperform univariate benchmarks and all other ML methods, including LASSO, bagging, and Bayesian VAR. Their research suggests that non-linear and non-parametric methods perform better, as inflation dynamics might change over time. This perspective motivates the application of neural networks, a highly non-linear and non-parametric ML method. Almosova and Andresen (2023) applied FNN and LSTM in a univariate framework, using lagged inflation as inputs and found slight improvements over AR. Paranhos (2025) applied multi-layer perceptions and LSTM in a data rich setting and found that LSTM systematically outperforms univariate benchmarks, especially at long forecasting horizons. This project builds on their work by applying FNN and LSTM models to US CPI inflation forecasting with extended sample and comparison benchmarks.

3 Data

This research uses data from the FRED-MD database constructed by McCracken and Ng (2016), which is a monthly macroeconomic database covering 126 series and 802 observations from March 1959 to December 2025. The forecasting target is the log difference of CPI all-item series. Other series are transformed to stationarity using transformation codes offered by the database. Series with more than 10% missing values are dropped, leaving 122 series. Among these, only one has approximately 5% missing values and the rest have less than 0.05%. The training sample is set to span from March 1959 to July 2012 (80% of observations), while the test sample is from August 2012 to December 2025.

Remaining missing values are imputed using scikit-learn's iterative imputer. It treats missing values of each variable as a target and estimate them through Bayesian ridge regression on other features. Given the high dimensionality and correlation among macroeconomic variables, principal component analysis (PCA) is applied to extract the top 10 principal components as exogenous predictor. Before PCA, all variables are standardized. After PCA, the predictors and the inflation series are min-max normalized to the range [0, 1], as Paranhos (2025) suggested that neural networks perform better when all features share the same range. All transformations are fitted on the training set and applied to the full sample to prevent data leakage.

4 Methodology

4.1 Forecasting Framework

To forecast the inflation h periods ahead, there are two frameworks (Bredahl Kock & Teräsvirta, 2016). The indirect forecasting starts with one-step-ahead forecasting, inserts the prediction as predictor for two-step-ahead forecasting, and iterates recursively to reach h -step-ahead prediction. In contrast, the direct forecasting estimates a separate model for each horizon. While indirect forecasting can be more efficient than direct forecasting, it requires assumptions on error terms and forecasting of covariates (Marcellino et al., 2005). Therefore, this research adopts direct forecasting:

$$y_{t+h} = G_h(z_t, x_t) + \mu_{t+h}$$

where y_{t+h} is the inflation rate h months ahead, $z_t = [y_t, y_{t-1}, \dots, y_{t-p}]'$ is a vector containing p lags of inflation, x_t is a vector of current and lagged covariates, and G_h is a horizon specific mapping. For each model, the mapping $G_h(\cdot)$ is estimated using the same specification and features but a different target corresponding to horizon h . The horizon h are set to 1, 3, 6, 12 and 24.

4.2 Models

The FNN adopted is a fully connected neural network with one hidden layer:

$$\hat{y}_{t+h} = \phi_0 + \sum_{n=1}^N \phi_n a\left(\sum_{p=0}^P w_{n,p} y_{t-p} + \sum_{k=1}^K \sum_{q=0}^Q \beta_{n,k,q} x_{k,t-q}\right)$$

where y_t denotes inflation at period t , $x_{k,t}$ denotes the k -th principal component at period t , N is the number of nodes in the neural network, P is the number of lags of inflation, Q is the number of lags of the principal components, and $a(\cdot)$ is the ReLU activation function.

The LSTM is a type of recurrent neural network (RNN). The RNN takes a sequence of the last P observations as input to predict the h -step-ahead observation. For each observation in the sequence, it encodes the observation into a hidden state that depends on both the current observation and the previous hidden state. It forms the prediction by applying a function to the final hidden state of the sequence. This feedback among hidden states allow RNN to remember distant observation when making prediction. However, RNN suffers from the vanishing gradient problem when the input sequence is long (Masini et al., 2023). LSTM solves this issue by introducing a cell state with gate mechanisms. The LSTM adopted can be expressed as:

$$\begin{aligned} f_t &= \text{Logit}(\beta_f X_t + w_f H_{t-1} + b_f), \quad i_t = \text{Logit}(\beta_i X_t + w_i H_{t-1} + b_i) \\ o_t &= \text{Logit}(\beta_o X_t + w_o H_{t-1} + b_o), \quad p_t = \text{Tanh}(\beta_p X_t + w_p H_{t-1} + b_p) \\ c_t &= (f_t \odot c_{t-1}) + (i_t \odot p_t), \quad H_t = o_t \odot \text{Tanh}(c_t), \quad \hat{y}_{t+h} = w_2 a[w_1 H_t + b_1] + b_2 \end{aligned}$$

The vector X_t denotes inflation and principal components at period t , H_t is the hidden state, C_t is the cell state carrying long-term information, and p_t is a scaled combination of X_t and the past hidden state. The vectors f_t , i_t and o_t take values between 0 and 1 and act as gating mechanisms. The f_t controls the proportion of C_{t-1} to retain and i_t controls the proportion of p_t to incorporate into C_t . The o_t controls the mapping from cell state C_t to hidden state H_t . Finally, the hidden state H_t passes through a one-layer fully connected neural network to form the prediction. Given a sequence of inputs $\{X_{t-p}, \dots, X_{t-1}, X_t\}$, the LSTM encodes a hidden state for each X , but only uses the last hidden state H_t to predict y_{t+h} . The length of the sequence is denoted as W , the window size.

Beyond neural networks, this research uses AR, autoregressive distributed lag (ARDL), LASSO, random forest, and gradient boosting machine (GBM) as benchmarks. Model parameters are estimated by minimizing mean squared error (MSE) in training set. For neural networks, parameters are estimated using Adam optimization algorithm, a variant of stochastic gradient descent, with a batch size of 128. To prevent over-fitting given the small sample size, 20% of hidden layer nodes are randomly dropped for each training batch.

4.3 Hyper-parameter Tuning

Hyper-parameters are selected using grid-search: for each set of hyper-parameters, cross-validation loss is evaluated, and the set with lowest cross-validation loss is selected. For time-series data, cross-validation does not rely on random splitting. Instead, the sample is split in chronological order. In this research, the training set is divided into 5 chronological folds. For each fold, the model is trained on observations up to that fold and evaluated on the subsequent fold. Cross-validation loss is computed by averaging MSE across folds. Table 1 reports the hyper-parameters considered for each model, the choices evaluated in grid-search, and the optimal values. Some hyper-parameters are fixed rather than grid-searched, as adding

dimensions would significantly increase computation time. For FNN, the number of inflation lags is set to 11, inspired by the optimal choice in AR. The number of principal component lags in FNN is set to 0, meaning only current period principal components are included. This choice is based on ARDL grid-search result, which suggests that 0 lag is optimal. For LSTM, epochs and learning rate are set as optimal values from FNN. For LASSO, random forest, and GBM, inflation and principal components lags are set to 11.

Table 1: Model Hyper-parameters

Model	Hyper-parameter	Choices	Optimal Value
FNN	Nodes	[8, 12, 16]	12
	Epochs	[100, 300]	100
	Learning Rate	[0.001, 0.0001]	0.001
LSTM	Nodes	[8, 12, 16]	12
	Hidden State Size	[8, 12, 16]	16
	Window Size	[6, 12]	12
AR	Lags	[3, 7, 11, 15, 19, 23]	11
ARDL	Inflation Lags	[3, 7, 11, 15, 19, 23]	7
	Principal Components Lags	[0, 3, 7, 11, 15, 19, 23]	0
LASSO	Alpha	[0.1, 0.02, 0.01, 0.002, 0.001]	0.01
Random Forest	Estimators	[100, 200, 500]	500
	Min Sample Split	[2, 4, 6]	4
GBM	Estimators	[100, 200, 500]	100
	Learning rate	[0.001, 0.005, 0.01, 0.05, 0.1]	0.01
	Max Depth	[-1, 3, 5]	5

5 Forecasting Performance

Paranhos (2025) noted that neural networks are highly sensible to initial values. Therefore, for each neural network model specification, 20 models with different initial values are estimated on the training set and ensemble to produce an average prediction. Tables 2 presents the forecasting performance of each model in terms of root MSE relative to AR across each horizon; it reveals several findings. First, the univariate AR model outperforms multivariate models including ARDL, LASSO, random forest, and GBM at almost every horizon, supporting the view of Stock and Watson (2010). Second, FNN achieves better performance relative to AR at short forecasting horizons (1 to 6), while LSTM outperforms AR at long horizons (3 to 24). This finding illustrates the advantage of neural networks for inflation forecasting in data-rich settings, especially compared to the weak performance of other ML methods. Third, FNN outperforms LSTM at 1-step-ahead horizon but performs worse at all longer horizons.

Table 2: Out-of-sample Forecasting RMSE Relative to AR

Model	1	3	6	12	24
FNN	93.09	97.95	98.62	106.50	101.03
LSTM	115.31	94.62	89.32	85.10	97.08
ARDL	98.59	109.61	105.66	114.73	107.74
LASSO	142.34	126.44	120.05	116.32	106.43
Random Forest	136.40	119.75	118.41	111.70	103.91
GBM	130.46	116.01	114.32	105.94	97.71

6 Discussion

Several limitations could be addressed in future work. First, grid search was applied to only a subset of hyper-parameters and selected optimal values only for one-step-ahead forecasting. Future research could include FNN lags and LSTM learning rate and epochs in the grid search and implement separate search for each forecasting horizon. Second, neural networks are highly sensitive to initial values and thus cross-validation loss for each hyper-parameter set could be evaluated using an ensemble of networks with different initial values rather than a single network.

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